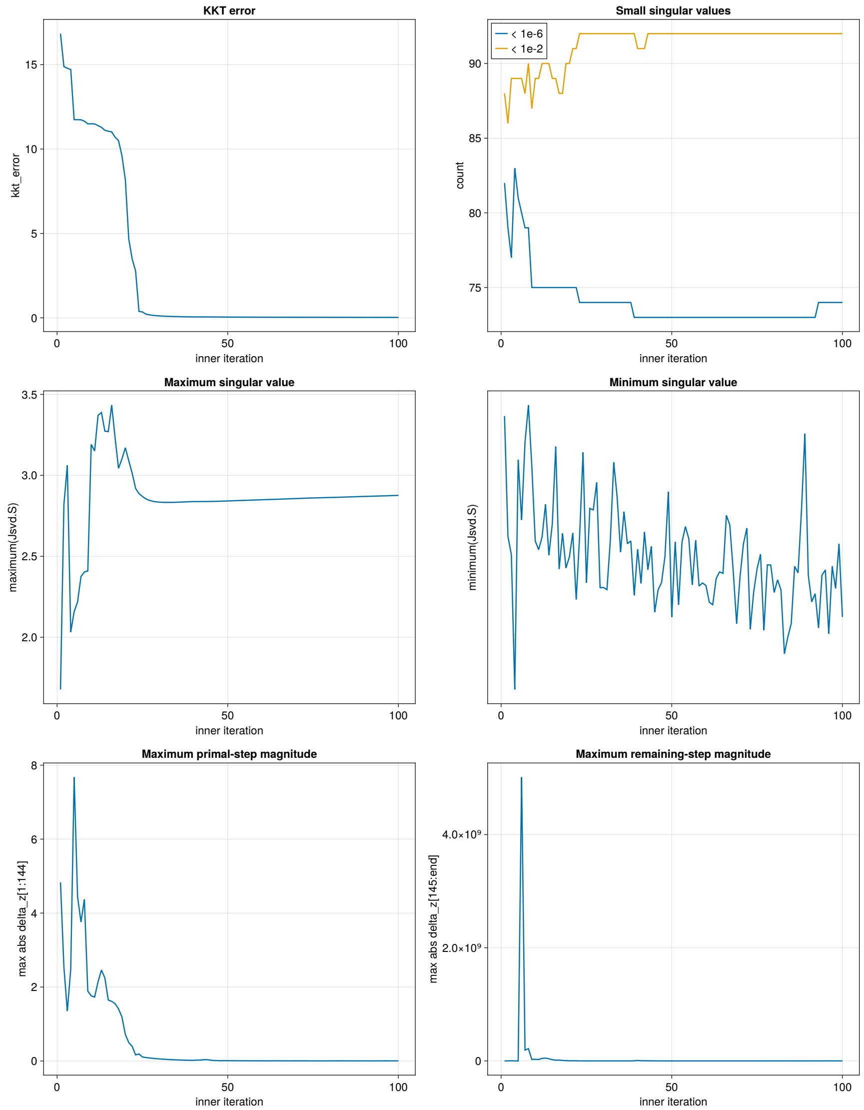
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
`kkt_error`	2.583087e-02	1.683624e+01	2.528381e+00	4.676754e-02
`count(Jsvd.S < 1e-6)`	7.300000e+01	8.300000e+01	7.408000e+01	7.300000e+01
`count(Jsvd.S < 1e-2)`	8.600000e+01	9.200000e+01	9.132000e+01	9.200000e+01
`maximum(Jsvd.S)`	1.677692e+00	3.433673e+00	2.855034e+00	2.855074e+00
`minimum(Jsvd.S)`	1.547308e-17	7.647105e-17	4.405166e-17	4.203232e-17
`maximum(abs.(delta_z[1:144]))`	1.735565e-03	7.674958e+00	5.420838e-01	7.528901e-03
`maximum(abs.(delta_z[145:end]))`	3.777701e+01	5.015157e+09	5.742621e+07	1.335161e+05

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max\lvert\delta z_{1:144}\rvert$	$\max\lvert\delta z_{145:}\rvert$
1	1.683624e+01	82	88	1.677692e+00	7.413368e-17	4.832770e+00	3.777701e+01
2	1.488129e+01	79	86	2.819098e+00	4.830340e-17	2.538178e+00	1.378903e+05
3	1.478788e+01	77	89	3.061255e+00	4.438921e-17	1.350837e+00	3.399021e+06
4	1.471419e+01	83	89	2.032031e+00	1.547308e-17	2.481602e+00	1.921799e+04
5	1.174229e+01	81	89	2.158214e+00	6.475722e-17	7.674958e+00	2.376324e+05
6	1.173907e+01	80	89	2.217966e+00	5.184522e-17	4.445559e+00	5.015157e+09
7	1.173448e+01	79	88	2.374387e+00	6.845880e-17	3.758306e+00	1.919554e+08
8	1.165037e+01	79	90	2.403446e+00	7.647105e-17	4.369700e+00	2.163756e+08
9	1.149629e+01	75	87	2.409276e+00	6.327901e-17	1.888646e+00	2.622952e+07
10	1.149570e+01	75	89	3.190885e+00	4.730961e-17	1.763281e+00	2.908726e+07
11	1.148894e+01	75	89	3.150992e+00	4.546539e-17	1.728019e+00	2.487820e+07
12	1.138541e+01	75	90	3.370132e+00	4.816754e-17	2.140554e+00	4.538882e+07
13	1.128834e+01	75	90	3.388028e+00	5.517491e-17	2.458555e+00	4.938068e+07
14	1.111164e+01	75	90	3.272713e+00	4.429906e-17	2.246115e+00	3.819933e+07
15	1.105837e+01	75	89	3.269910e+00	5.105791e-17	1.648730e+00	2.172726e+07
16	1.101344e+01	75	89	3.433673e+00	6.758070e-17	1.612039e+00	1.394348e+07
17	1.070457e+01	75	88	3.230883e+00	4.132446e-17	1.549410e+00	1.419725e+07
18	1.049993e+01	75	88	3.043298e+00	4.893805e-17	1.410343e+00	8.855515e+06
19	9.610508e+00	75	90	3.101554e+00	4.153237e-17	1.192149e+00	5.553209e+06
20	8.183611e+00	75	90	3.170307e+00	4.389966e-17	7.179209e-01	2.795846e+06
21	4.674124e+00	75	91	3.090713e+00	4.902071e-17	4.980632e-01	2.692090e+06
22	3.486972e+00	75	91	3.013637e+00	3.478208e-17	3.955184e-01	2.034906e+06
23	2.786476e+00	74	92	2.919656e+00	4.841977e-17	1.648604e-01	8.536819e+05
24	3.793284e-01	74	92	2.886899e+00	6.632625e-17	1.897361e-01	4.433252e+05
25	3.499516e-01	74	92	2.869790e+00	3.839733e-17	1.081248e-01	2.569796e+05
26	2.239834e-01	74	92	2.854960e+00	5.430554e-17	9.396640e-02	1.756625e+05
27	1.781438e-01	74	92	2.845738e+00	5.400180e-17	8.261518e-02	1.426098e+05
28	1.486805e-01	74	92	2.839984e+00	5.988195e-17	7.255860e-02	1.202590e+05
29	1.287874e-01	74	92	2.836483e+00	3.729414e-17	6.377145e-02	1.036889e+05
30	1.141964e-01	74	92	2.834449e+00	3.734735e-17	5.610856e-02	9.080804e+04
31	1.028538e-01	74	92	2.833386e+00	3.687248e-17	4.942034e-02	8.043746e+04
32	9.367656e-02	74	92	2.832977e+00	4.748458e-17	4.357533e-02	7.186126e+04
33	8.605481e-02	74	92	2.833015e+00	6.419521e-17	3.846114e-02	6.462123e+04
34	7.961642e-02	74	92	2.833363e+00	5.668798e-17	3.398178e-02	5.841039e+04
35	7.411584e-02	74	92	2.833927e+00	4.493427e-17	3.005494e-02	5.301470e+04
36	6.937926e-02	74	92	2.834645e+00	5.360142e-17	2.660969e-02	4.827967e+04
37	6.527649e-02	74	92	2.835470e+00	4.674835e-17	2.358472e-02	4.409036e+04
38	6.170567e-02	74	92	2.836370e+00	4.732277e-17	2.092690e-02	4.035902e+04
39	5.858460e-02	73	92	2.837323e+00	3.565688e-17	1.859018e-02	2.385680e+06
40	5.618256e-02	73	91	2.837809e+00	4.556365e-17	1.757071e-02	7.880176e+06
41	5.578513e-02	73	91	2.837874e+00	3.822313e-17	2.589293e-02	3.352657e+06
42	5.535223e-02	73	91	2.838067e+00	4.924418e-17	2.536958e-02	2.529587e+06
43	5.469769e-02	73	92	2.838251e+00	4.117295e-17	3.590411e-02	2.178256e+06
44	5.442473e-02	73	92	2.838402e+00	4.615740e-17	3.539980e-02	1.707957e+06
45	5.441670e-02	73	92	2.838705e+00	3.206142e-17	2.198335e-02	6.927542e+05
46	5.397189e-02	73	92	2.839045e+00	3.686393e-17	1.302597e-02	2.517125e+05
47	5.148818e-02	73	92	2.839820e+00	3.838558e-17	1.182114e-02	1.006834e+05
48	5.049352e-02	73	92	2.840612e+00	4.398371e-17	6.429061e-03	2.573615e+04
49	4.804533e-02	73	92	2.841052e+00	5.788121e-17	9.302916e-03	3.831784e+04
50	4.735201e-02	73	92	2.841798e+00	3.102012e-17	8.534556e-03	2.737817e+04
51	4.618307e-02	73	92	2.842539e+00	4.712524e-17	7.842976e-03	2.172399e+04
52	4.514500e-02	73	92	2.843284e+00	3.362503e-17	7.214825e-03	1.782448e+04
53	4.418816e-02	73	92	2.844030e+00	4.707870e-17	6.643964e-03	1.544719e+04
54	4.330132e-02	73	92	2.844775e+00	5.041848e-17	6.124834e-03	1.439604e+04
55	4.247586e-02	73	92	2.845518e+00	4.776599e-17	5.652447e-03	1.344800e+04
56	4.170495e-02	73	92	2.846259e+00	3.790786e-17	5.222324e-03	1.257914e+04
57	4.098287e-02	73	92	2.846996e+00	4.750175e-17	4.830440e-03	1.177278e+04
58	4.030467e-02	73	92	2.847728e+00	3.766408e-17	4.473177e-03	1.101667e+04
59	3.966605e-02	73	92	2.848455e+00	3.830744e-17	4.147277e-03	1.030140e+04
60	3.906319e-02	73	92	2.849177e+00	3.778406e-17	3.849803e-03	9.619412e+03
61	3.849277e-02	73	92	2.849892e+00	3.420012e-17	3.696466e-03	8.964335e+03
62	3.795181e-02	73	92	2.850600e+00	3.364000e-17	3.604977e-03	8.330554e+03

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max\lvert\delta z_{1:144}\rvert$	$\max\lvert\delta z_{145:}\rvert$
63	3.743772e-02	73	92	2.851302e+00	3.925952e-17	3.517268e-03	7.712855e+03
64	3.694818e-02	73	92	2.851997e+00	4.068699e-17	5.403179e-03	1.066780e+04
65	3.662442e-02	73	92	2.852542e+00	4.038312e-17	5.303505e-03	9.967763e+03
66	3.660402e-02	73	92	2.853631e+00	5.282172e-17	2.942483e-03	5.303375e+03
67	3.545363e-02	73	92	2.854227e+00	5.076877e-17	4.556126e-03	7.035345e+03
68	3.529927e-02	73	92	2.855187e+00	4.091503e-17	2.539670e-03	3.638374e+03
69	3.454223e-02	73	92	2.855716e+00	2.961141e-17	3.942797e-03	4.604125e+03
70	3.431642e-02	73	92	2.856565e+00	3.965843e-17	3.828102e-03	4.324885e+03
71	3.385249e-02	73	92	2.857395e+00	4.672111e-17	3.724015e-03	5.651401e+03
72	3.342904e-02	73	92	2.858212e+00	5.007757e-17	3.625202e-03	8.933120e+03
73	3.306883e-02	73	92	2.859016e+00	2.838014e-17	3.530865e-03	1.606885e+04
74	3.279525e-02	73	92	2.859807e+00	3.639044e-17	1.981756e-03	1.849768e+04
75	3.217187e-02	73	92	2.860252e+00	4.138734e-17	3.087622e-03	3.104337e+04
76	3.211198e-02	73	92	2.860958e+00	4.449414e-17	1.735565e-03	2.812517e+04
77	3.160444e-02	73	92	2.861352e+00	2.817979e-17	2.707761e-03	4.272675e+04
78	3.144243e-02	73	92	2.861979e+00	4.219470e-17	2.649898e-03	5.416964e+04
79	3.113093e-02	73	92	2.862596e+00	4.217459e-17	2.596251e-03	6.250405e+04
80	3.083272e-02	73	92	2.863203e+00	3.628926e-17	2.544563e-03	7.099096e+04
81	3.054963e-02	73	92	2.863803e+00	3.898921e-17	2.494618e-03	8.003076e+04
82	3.027827e-02	73	92	2.864395e+00	3.682380e-17	3.854892e-03	1.433025e+05
83	3.009737e-02	73	92	2.864856e+00	2.311702e-17	3.796999e-03	1.578520e+05
84	3.005281e-02	73	92	2.865771e+00	2.671669e-17	2.119493e-03	1.167356e+05
85	2.943155e-02	73	92	2.866278e+00	2.958374e-17	3.295502e-03	1.829246e+05
86	2.930090e-02	73	92	2.867082e+00	4.189005e-17	1.846385e-03	1.291419e+05
87	2.889567e-02	73	92	2.867530e+00	4.048347e-17	2.876408e-03	2.045666e+05
88	2.872725e-02	73	92	2.868239e+00	5.392170e-17	2.806689e-03	2.388499e+05
89	2.844041e-02	73	92	2.868931e+00	7.031038e-17	2.742740e-03	2.682027e+05
90	2.816180e-02	73	92	2.869612e+00	3.997833e-17	2.681400e-03	2.994199e+05
91	2.789231e-02	73	92	2.870281e+00	3.425819e-17	2.622379e-03	3.331987e+05
92	2.763142e-02	73	92	2.870939e+00	3.601554e-17	2.565543e-03	3.697813e+05
93	2.737862e-02	74	92	2.871585e+00	2.867285e-17	2.510780e-03	4.093947e+05
94	2.713341e-02	74	92	2.872221e+00	3.998799e-17	2.457985e-03	4.522873e+05
95	2.689540e-02	74	92	2.872844e+00	4.109329e-17	2.407052e-03	4.589350e+05
96	2.666300e-02	74	92	2.873457e+00	2.745271e-17	3.715385e-03	6.516749e+05
97	2.650528e-02	74	92	2.873932e+00	4.188045e-17	3.656469e-03	5.996804e+05
98	2.648112e-02	74	92	2.874874e+00	3.715273e-17	2.036975e-03	3.024831e+05
99	2.592678e-02	74	92	2.875396e+00	4.670477e-17	3.166088e-03	4.058025e+05
100	2.583087e-02	74	92	2.876219e+00	3.102192e-17	1.770811e-03	2.045389e+05